IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Title – Smart Study Generator Agent-Problem statement 13

Domain of Project – Education

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Problem Statement -13

Smart Study Generator Agent

The Challenge

Students today face an overwhelming flood of information—lecture notes, textbooks, research papers, YouTube tutorials, and online resources. This content is scattered, unorganized, and often not personalized to their learning pace or style. As a result, students struggle with finding the right study material, summarizing key concepts, and staying exam-ready. Traditional study tools only provide static notes or flashcards, failing to adapt dynamically to student needs. The challenge is to move beyond static resources toward an intelligent, adaptive study companion.

The Objective

Build an agentic AI-powered Smart Study Generator that personalizes and streamlines learning for students. The solution should:

Fuse Learning Resources: Aggregate content from class notes, textbooks, research articles, and online tutorials into structured, easy-to-digest study material.

Enable Multimodal Input: Allow students to upload notes, voice queries, or images (e.g., textbook pages), and generate concise summaries, flashcards, or concept maps.

Create a Predictive & Agentic Layer: Recommend personalized study plans, highlight important topics based on syllabus or past exams, and provide predictive alerts (e.g., “Focus on Topic X—frequently asked in exams”).

Visualize Study Progress: Offer a real-time dashboard showing learning milestones, weak areas, practice quizzes, and performance forecasts.

Tech Stack (Mandatory)

IBM Langflow / IBM Orchestrate + IBM Models

Give a prompt for ibm bob

Proposed Solution -

The Smart Study Generator Agent is an AI-powered platform that converts learning resources into personalized study materials.

Key capabilities:

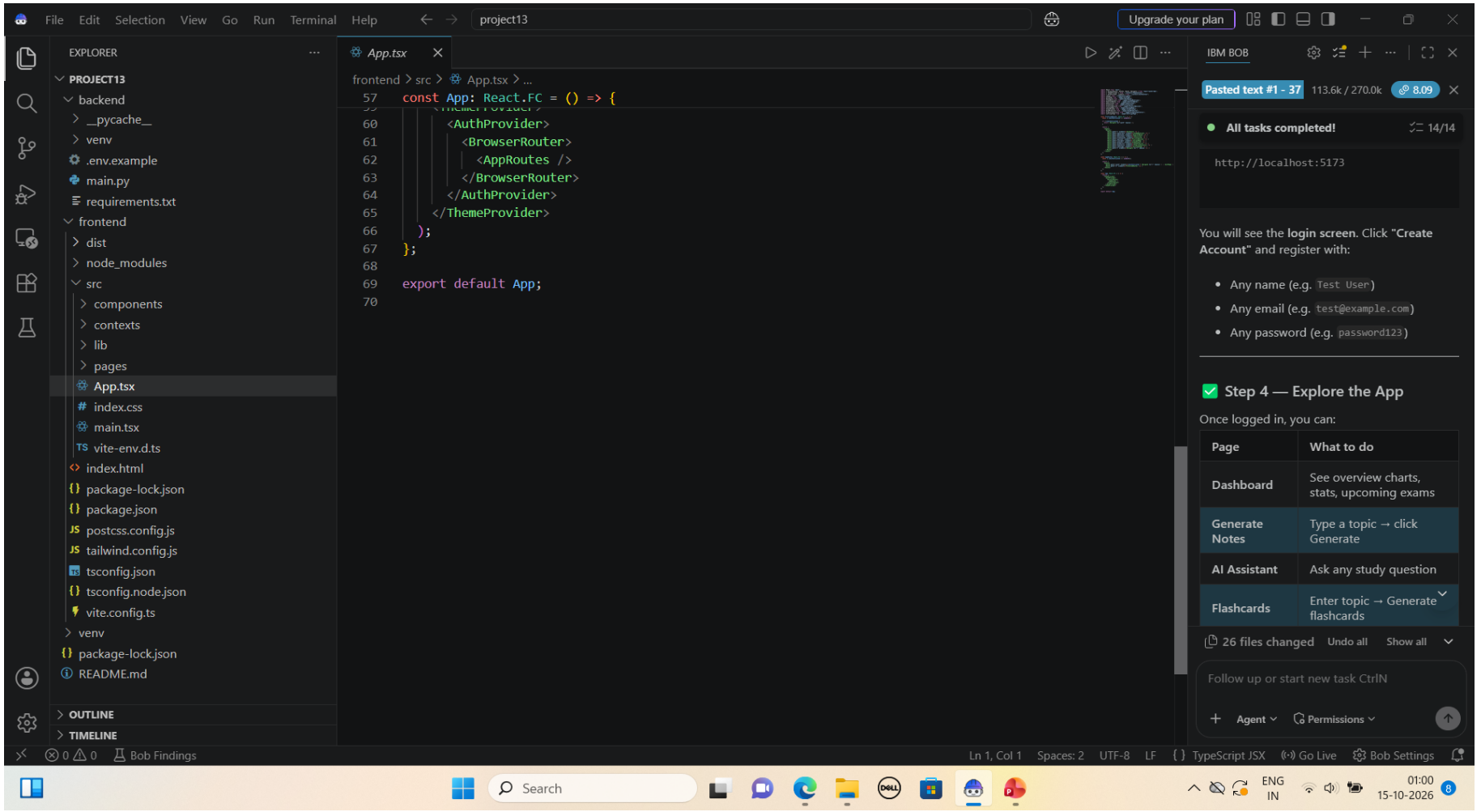
- Upload PDFs, lecture notes, and textbook images.
- Generate summaries and structured study notes.
- Automatically create flashcards, quizzes, and concept maps.
- Identify weak topics through quiz performance.
- Recommend personalized daily and weekly study plans.
- Analyze previous examination papers to identify frequently asked topics.
- Display learning progress through an interactive dashboard.

Technology Used

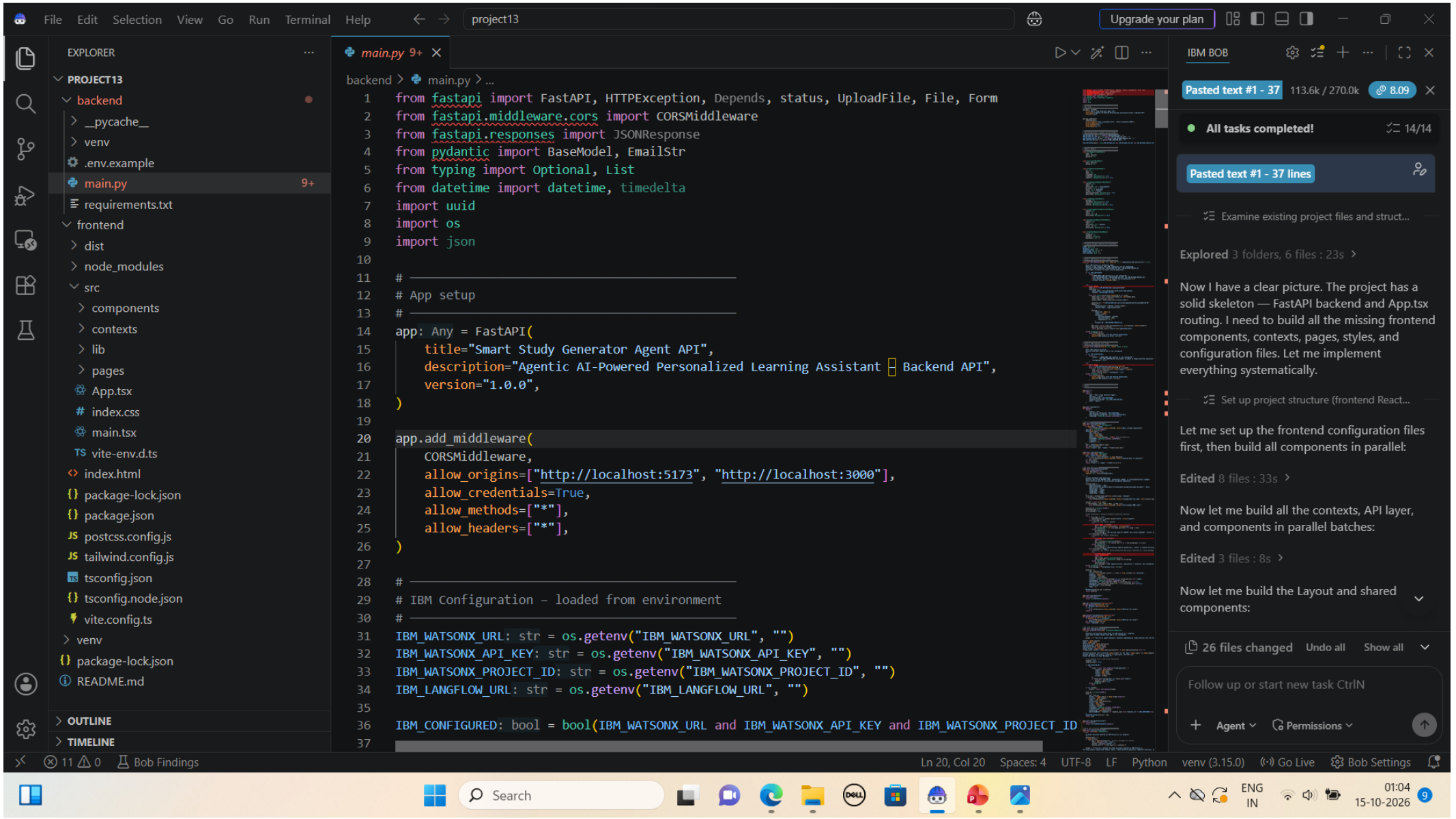
- IBM Bob platform -For designing and orchestrating multi-agent workflows visually
- Frontend: Provides an interactive interface for students.
- Backend: Manages requests, data, and AI integrations.
- Document processing: Extracts text from uploaded learning materials.
- IBM Langflow: Connects and orchestrates AI workflows.
- IBM watsonx.ai: Powers summarization, explanations, and content generation.
- IBM Orchestrate: Supports suitable task coordination and automation.
- Database: Stores study materials, quiz results, and learning progress.

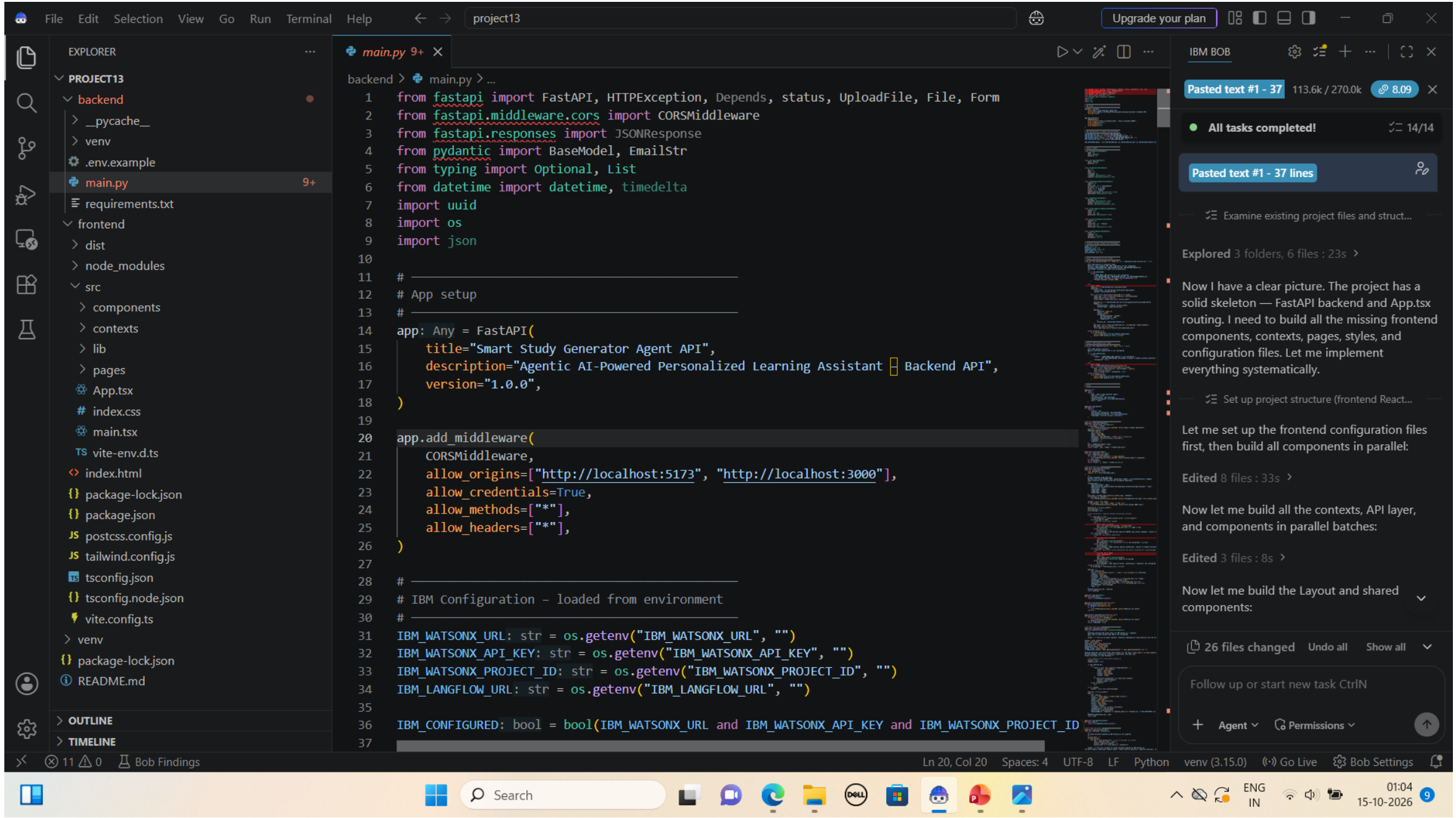
Required Files created using IBM bob -

1. App.py
2. Requirements.txt
3. Style.css
4. .env
5. Index.html



IBM Bob Screenshots:

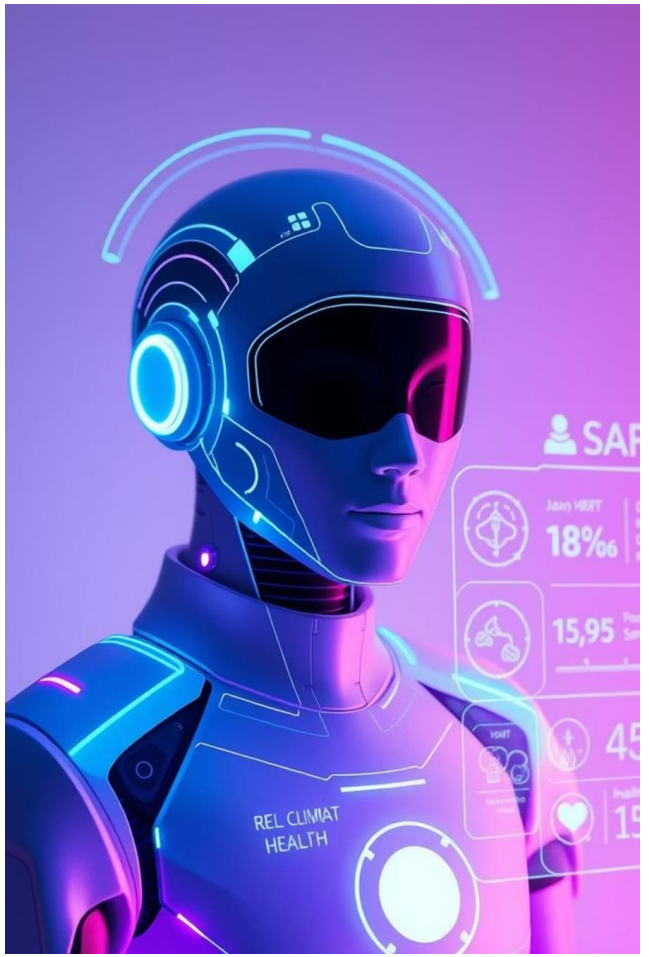



File Edit Selection View Go Run Terminal Help
project13
Upgrade your plan
EXPLORER
PROJECT13
backend
__pycache__
venv
.env.example
main.py 9+
requirements.txt
frontend
dist
node_modules
src
components
contexts
lib
pages
App.tsx
index.css
main.tsx
vite-env.d.ts
index.html
package-lock.json
package.json
postcss.config.js
tailwind.config.js
tsconfig.json
tsconfig.node.json
vite.config.ts
venv
package-lock.json
README.md
OUTLINE
TIMELINE
main.py 9+
backend > main.py > ...
from fastapi import FastAPI, HTTPException, Depends, status, UploadFile, File, Form
from fastapi.middleware.cors import CORSMiddleware
from fastapi.responses import JSONResponse
from pydantic import BaseModel, EmailStr
from typing import Optional, List
from datetime import datetime, timedelta
import uuid
import os
import json
App setup
app: Any = FastAPI(
title="Smart Study Generator Agent API",
description="Agentic AI-Powered Personalized Learning Assistant - Backend API",
version="1.0.0",
)
app.add_middleware(
CORSMiddleware,
allow_origins=["http://localhost:5173", "http://localhost:3000"],
allow_credentials=True,
allow_methods=["*"],
allow_headers=["*"],
)
IBM Configuration - loaded from environment
IBM_WATSONX_URL: str = os.getenv("IBM_WATSONX_URL", "")
IBM_WATSONX_API_KEY: str = os.getenv("IBM_WATSONX_API_KEY", "")
IBM_WATSONX_PROJECT_ID: str = os.getenv("IBM_WATSONX_PROJECT_ID", "")
IBM_LANGFLOW_URL: str = os.getenv("IBM_LANGFLOW_URL", "")
IBM_CONFIGURED: bool = bool(IBM_WATSONX_URL and IBM_WATSONX_API_KEY and IBM_WATSONX_PROJECT_ID
IBM BOB
Pasted text #1 - 37
113.6k / 270.0k
8.09
All tasks completed!
14/14
Pasted text #1 - 37 lines
Examine existing project files and struct...
Explored 3 folders, 6 files : 23s
Now I have a clear picture. The project has a solid skeleton — FastAPI backend and App.tsx routing. I need to build all the missing frontend components, contexts, pages, styles, and configuration files. Let me implement everything systematically.
Set up project structure (frontend React...
Let me set up the frontend configuration files first, then build all components in parallel:
Edited 8 files : 33s
Now let me build all the contexts, API layer, and components in parallel batches:
Edited 3 files : 8s
Now let me build the Layout and shared components:
26 files changed
Undo all
Show all
Follow up or start new task CtrlN
Agent
Permissions
Ln 20, Col 20
Spaces: 4
UTF-8
LF
Python
venv (3.15.0)
Go Live
Bob Settings
Bob Findings
11 0
Search
ENG
IN
01:04
15-10-2026

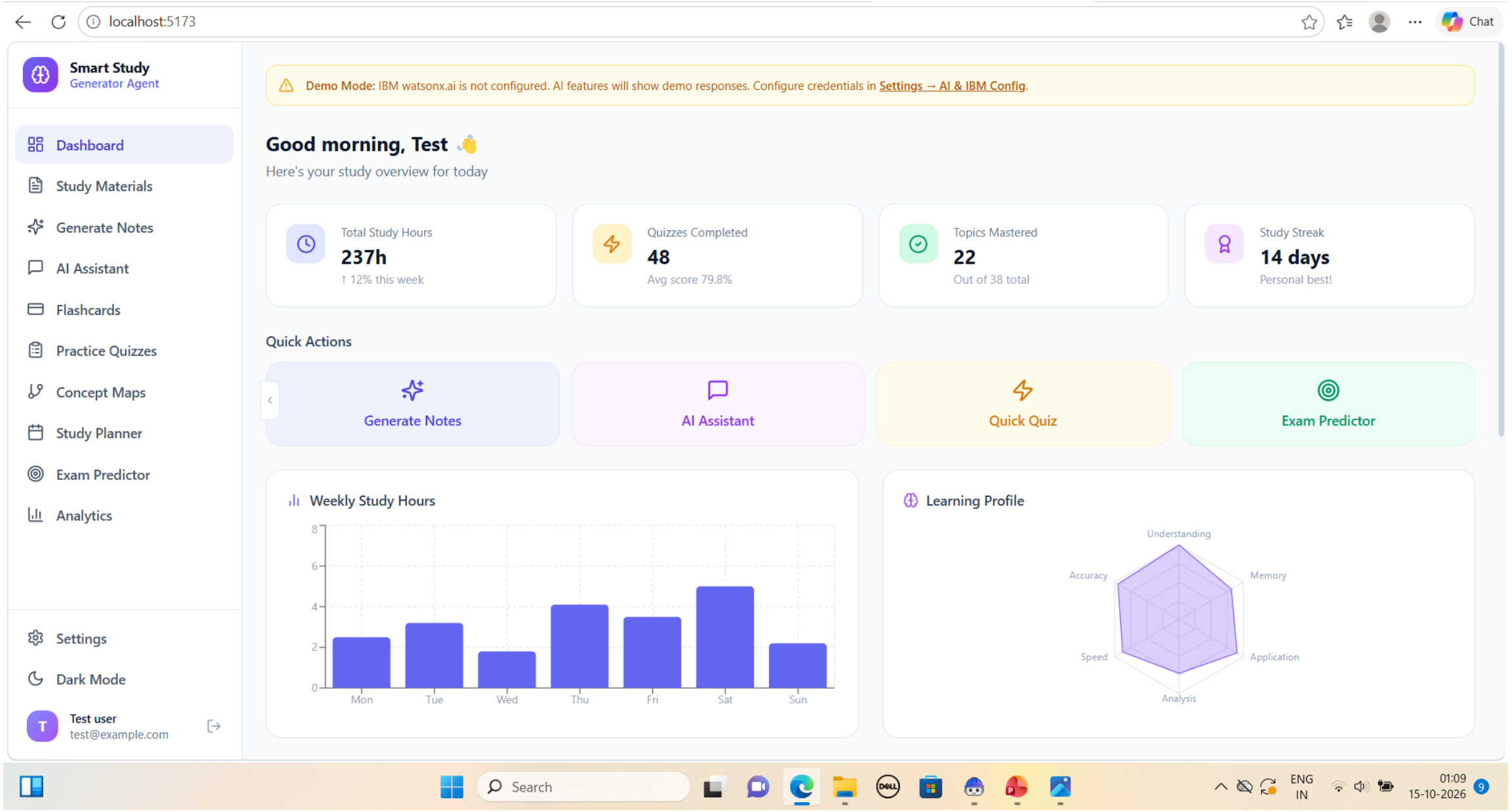
Role of Agentic AI in the solution

The system uses specialized AI workflows to provide personalized learning assistance.
Workflow:

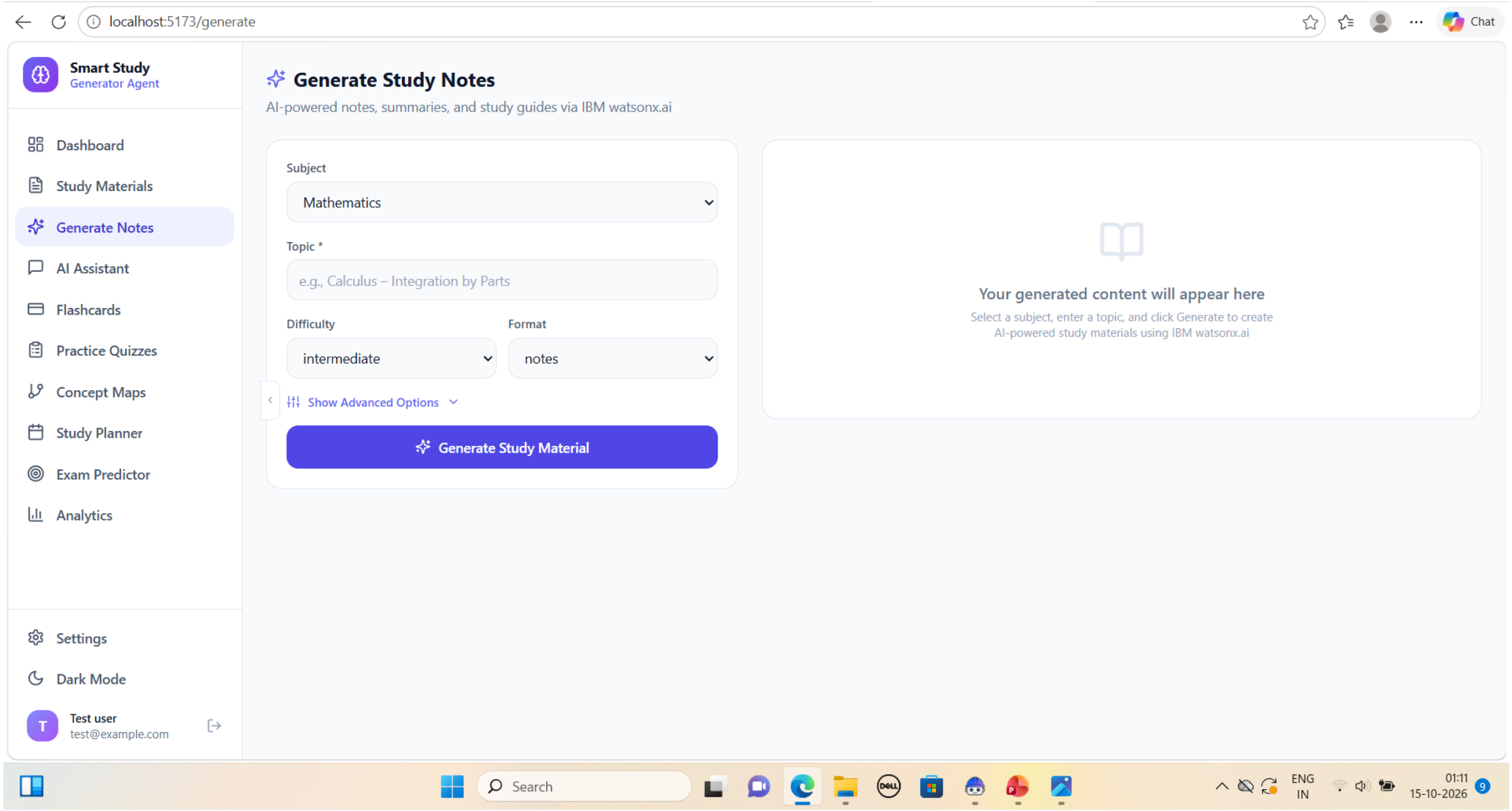
- Input Collection: Student uploads study materials and specifies a learning goal.
- Content Analysis: The system extracts text and identifies important concepts.
- Task Planning: The AI coordinator determines which workflow is required.
- Content Generation: AI generates notes, flashcards, or practice quizzes.
- Performance Evaluation: Quiz results are analyzed to identify weak topics.
- Adaptive Recommendation: The system suggests revision materials and study activities.
- Progress Tracking: Student performance is updated on the dashboard.



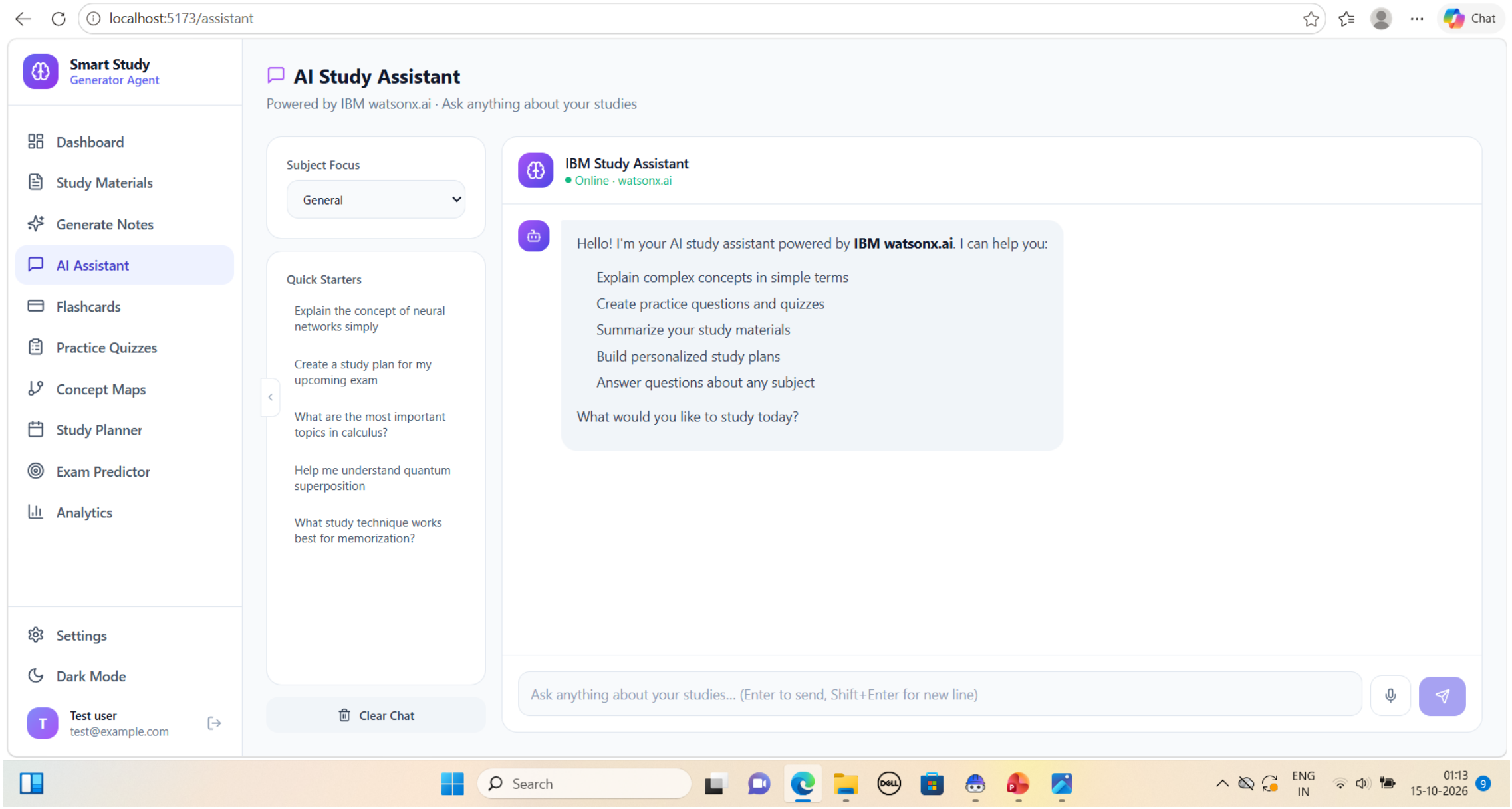
Project Output Screenshot



Project Output Screenshot



Project Output Screenshot



Novelty and Uniqueness

The proposed solution goes beyond conventional note-making and flashcard applications.
Innovative aspects:

- Agentic Learning: AI workflows select and execute tasks based on student needs.
- Multimodal Learning: Students can provide documents, images, and voice queries.
- Adaptive Study Planning: Recommendations change according to performance and deadlines.
- Weakness-Based Learning: The system focuses on concepts where students need improvement.
- Evidence-Based Exam Analysis: Previous papers are analyzed to identify recurring topics.
- Automated Revision Loop: Weak topics trigger targeted notes, practice questions, and follow-up assessments.
- Unified Learning Platform: Multiple study tools are integrated into a single application.

Git Hub Link

https://github.com/Prathvi1506/Smart-Study-Generator-Agent.git

Future Scope

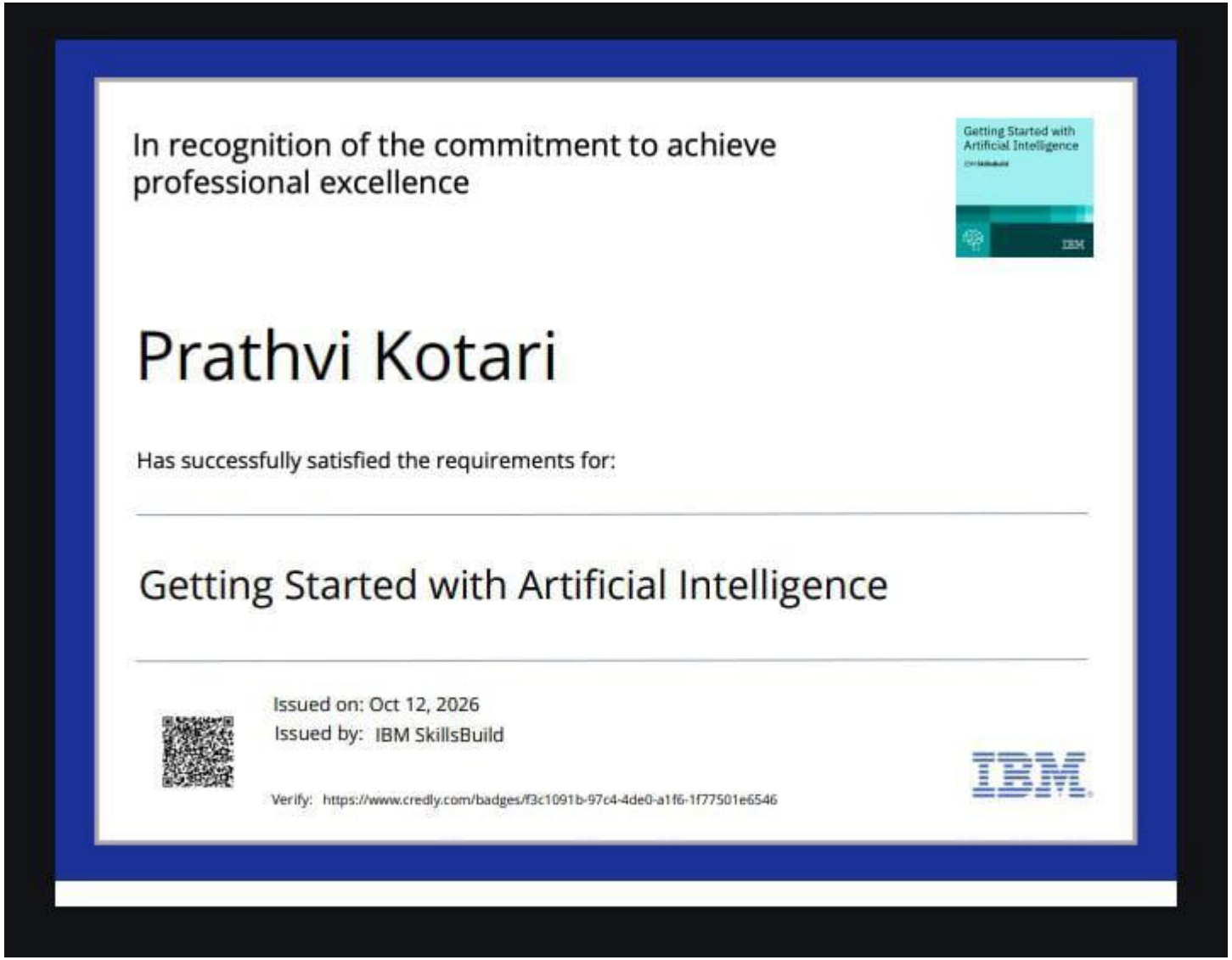
- Reduce the time required to prepare study notes.
- Make scattered learning resources easier to organize.
- Support personalized and self-paced learning.
- Help students identify weak topics earlier.
- Provide structured revision and practice opportunities.
- Improve visibility into learning progress.
- Help students prioritize examination preparation.

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Certificates Earned:

Please add your credly certificate



Certificates Earned:

Add your IBM BOB Certificate



Thank You!

Thank you for your time and interest.